

Risk Across Tails and Timescales

Does tail dependence change with the investment horizon,
and what does ignoring it do to measured risk?

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Summary

We look at a five-asset portfolio (US equities, tech stocks, long-term Treasuries, gold, and high-yield credit) from 12 April 2007 to 30 June 2026 (4,835 trading days). Using wavelets, we break daily returns into different time horizons, ranging from a single day up to about three months. We check how often assets crash together into their worst 5% across these horizons, and we run a rolling backtest to see what this means for Value-at-Risk (VaR).

- **Does tail dependence change with the horizon?** Across 30 pair-horizon tests (6 pairs across 5 multi-day horizons), only one has a 90% bootstrap interval for the change that excludes zero: equities and high-yield credit crash together much more over longer horizons. Their joint crash probability rises from 0.53 at one day to 0.77 beyond 64 days (with a 90% confidence interval of +0.10 to +0.37, holding up across 63-, 126-, and 252-day bootstrap blocks). Since this is the only pair out of 30 where the interval stays clear of zero, we treat it as an interesting lead rather than blanket proof across all markets. For equities and Treasuries, joint crashes rise from 0.05 to 0.22 (but the confidence interval crosses zero); for gold, it moves from 0.14 to 0.07 (interval also crosses zero); and comparing the portfolio's growth assets directly against its hedges shows no statistically significant change at any horizon.
- **What does ignoring it do to measured risk?** Fitting a copula to the actual 20-day horizon gives 5.0% higher VaR on average than a standard daily copula (and over 10.9% higher on one day in ten), reflecting the joint impact of horizon correlation and tail clustering combined. But in an out-of-sample backtest, the evidence favoring this higher risk estimate is weak: the Diebold-Mariano test cannot tell the models apart ($p = 0.32$ at 20 days, and at 5 days the simple daily $\sqrt{h}$ model actually scores best). The common $\sqrt{h}$ rule hides this entire issue because it inflates 20-day VaR by even more (8.7% too high on average).
- **Our recommendation.** For positions held 20 trading days or longer, stop multiplying 1-day VaR by $\sqrt{h}$. Instead, project volatility over the full horizon using GARCH and pair it with a copula fitted to that specific time horizon. In our out-of-sample test, this model achieved the best score at 20 days and saved 4.7% in capital on average compared to the $\sqrt{h}$ rule, without causing excess breaches (2 breaches in 191 windows; see Section 4).

1. Question, data and portfolio

Most risk models are built on daily data. But in the real world, investment desks hold positions over weeks or months. If the way assets crash together changes across different timescales, a model based purely on daily returns will misjudge long-term crash risk, even if its day-to-day volatility forecasts are accurate. In this report, we investigate whether this horizon shift actually happens and how much it affects measured portfolio risk.

We analyze five highly liquid ETFs, each representing a distinct source of market risk, with weights typical of a balanced institutional portfolio: SPY (S&P 500, 30%), QQQ (Nasdaq 100, 20%), TLT (20+ year Treasuries, 25%), GLD (gold, 15%), and HYG (high-yield corporate credit, 10%). We group SPY, QQQ, and HYG into a *risky sleeve*, and TLT and GLD into a *hedge sleeve*. This simple two-sleeve view gets straight to the practical question a portfolio manager cares about: do our hedges still protect us in a crash when we hold them for longer periods?

Price data were obtained from Yahoo Finance using adjusted prices; the exact dataset used in the analysis is included in the submission. All prices are daily dividend-adjusted closes. Our sample begins on the first day HYG started trading, so it covers major historical shocks: the 2008 financial crisis, the 2011 US credit downgrade, the COVID-19 crash in 2020, the 2022 inflation and rate hike shock, and the early 2025 tariff selloff. We work with log returns. We calculate portfolio returns as the fixed-weight sum of asset log returns ($r_{p,t} = \sum_i w_i r_{i,t}$), which is the standard daily approximation used by risk desks. All data is stored in the repository so the analysis runs without needing any external downloads.

2. Method

2.1. Horizon views from the wavelet transform

The maximal overlap discrete wavelet transform (MODWT) separates each return series r_t into detail components $D_1, \dots, D_J$ and a smooth trend S_J that sum back to the original series exactly: $r_t = \sum_{j=1}^J D_{j,t} + S_{J,t}$ [8]. Each detail level D_j captures price swings that occur over roughly 2^j to 2^{j+1} days. We set $J = 5$, which reaches past 64 trading days (roughly one calendar quarter), using the standard sym8 filter. Figure 1 shows this decomposition for SPY and TLT.

An isolated frequency band on its own does not represent an actual investment holding period. A trader holding an asset for 20 days experiences all cumulative market movements lasting 20 days or longer, not just a single narrow frequency slice. We therefore reconstruct cumulative *horizon views* by adding slower components back together:

$$V_t^{(0)} = r_t, \quad V_t^{(j)} = \sum_{k=j+1}^J D_{k,t} + S_{J,t}, \quad j = 1, \dots, J, \quad (1)$$

where view j retains all market swings lasting longer than 2^{j+1} days. Why use wavelets instead of simple non-overlapping h -day returns? If you take non-overlapping 64-day returns across our sample, you end up with fewer than 75 data points. That is far too small a sample to reliably fit a copula. Wavelet filtering strips away high-frequency noise while keeping all 4,835 daily observations, giving us enough data to estimate dependence accurately at every horizon.

Most market volatility happens at short timescales: on average, 79.6% of each asset's variance sits in D_1 and D_2 (movements under 8 days), while only 5.2% sits in the slow components D_5 and S_5 (Table 1). This means a standard copula fitted to daily data is almost entirely shaped by day-to-day noise and may not adequately represent dependence at longer investment horizons.

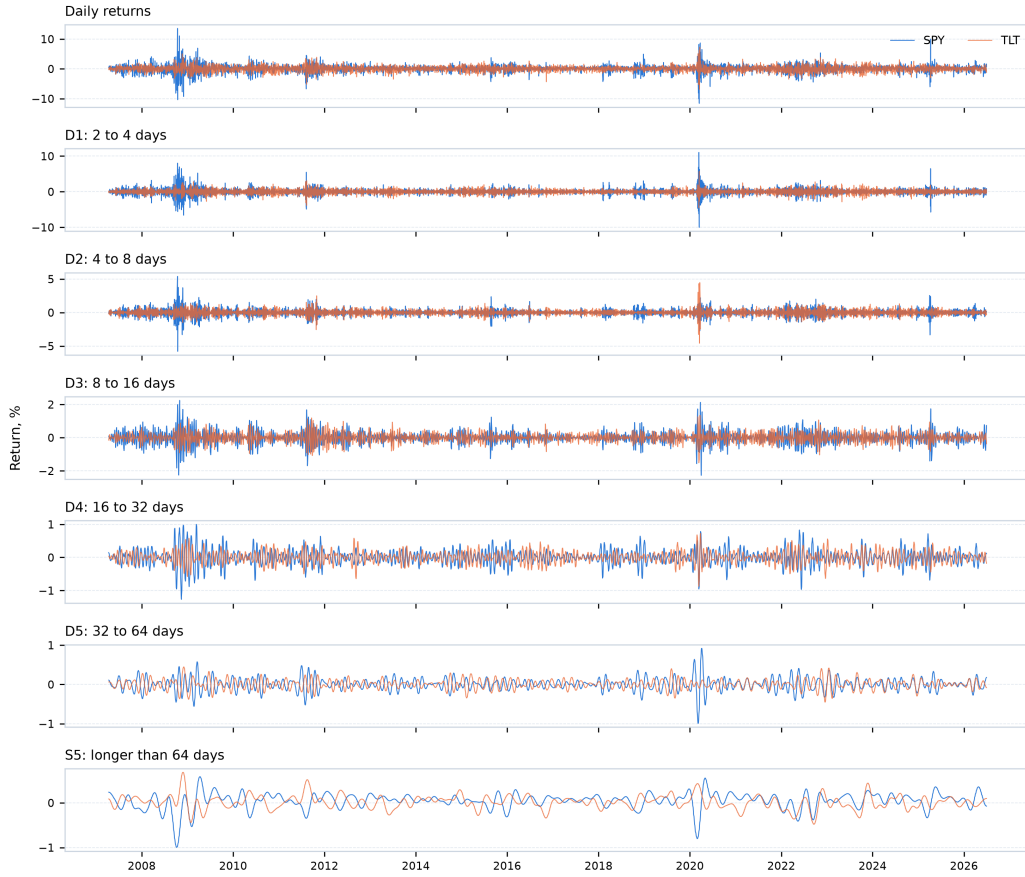


Figure 1: MODWT multiresolution analysis of SPY and TLT daily returns. The details and the smooth add back to the returns to machine precision.

Scale	SPY	QQQ	TLT	GLD	HYG
D1	62.1	60.7	56.3	55.2	54.5
D2	19.5	20.2	24.1	22.6	22.9
D3	9.9	10.2	10.5	11.5	11.9
D4	3.9	4.1	4.1	5.3	4.4
D5	2.0	2.1	2.0	2.6	3.0
S5	2.5	2.7	3.0	2.8	3.4

Table 1: Share of each asset's return variance in each MODWT scale (%).

2.2. Measuring tail dependence at each horizon

To measure how assets move together without getting distorted by individual asset distributions, we convert each series into ranks: $u_{i,t} = \text{rank}(x_{i,t}) / (n + 1)$. Ranks strip away the individual marginal distributions and map the data uniformly between 0 and 1, which is exactly what a copula requires. Our main metric is the lower-tail joint crash rate (co-exceedance) at level q :

$$\hat{\lambda}_L(q) = \frac{1}{q} \Pr(U_1 \leq q, U_2 \leq q), \quad (2)$$

which simply measures: on days when Asset 1 has one of its worst $q\%$ outcomes, how often is Asset 2 also having one of its worst $q\%$ outcomes? Throughout, we use 5% lower-tail co-exceedance as a finite-threshold proxy for tail dependence. If the two assets were completely independent, this would equal q . If they always crashed together in lockstep, it would equal 1. We look at $q = 5\%$, which gives us roughly 240 observations in each marginal tail at every horizon. We define the upper tail (joint rallies) the exact same way.

Linear correlation already causes some co-movement on its own. To see whether assets crash together more than pure correlation would suggest, we compare every estimate against a benchmark Gaussian copula that shares the exact same linear correlation ρ : $\Phi_2(\Phi^{-1}(q), \Phi^{-1}(q); \rho)/q$. Any points that rise clearly above this benchmark show genuine tail risk that simple correlation misses.

Because the smoothed wavelet views are rolling series, neighboring days are strongly correlated. Treating them as independent would make our confidence intervals unrealistically narrow. To handle this serial correlation properly, we use a moving block bootstrap with blocks of 126 days (about six months, long enough to capture several full cycles of our slowest view) and 500 resamples. Resampling the exact same blocks across all views lets us calculate reliable confidence intervals for the change between daily and multi-day horizons.

2.3. Copula family

For every asset pair and horizon view, we fit five standard copulas (Gaussian, Student- t , Clayton, Gumbel, and Frank) using maximum likelihood and pick the best fit using the Bayesian Information Criterion (BIC). The Student- t copula wins 40 of the 42 matchups, so we use it for all our VaR models below. Because the Student- t copula has symmetric tails, its job in our risk model is to capture the overall strength of tail dependence at each horizon. Any difference between crash risk and rally risk is evaluated directly from our empirical crash measurements.

2.4. VaR models

To isolate the effect of horizon dependence, all our copula models use identical marginal distributions. In our rolling backtest, each asset's daily volatility is estimated with a GJR-GARCH(1,1) model with Student- t innovations [4], using a constant mean to keep daily expected returns from compounding into unrealistic trends over longer holding periods. We model the standardized residuals with an empirical distribution in the body and generalized Pareto tails for the top and bottom 10% [6]. Over an h -day horizon, the forecasted variance is simply the sum of daily GARCH variance forecasts for days $1, \dots, h$, which naturally accounts for volatility mean reversion over time.

To calculate multi-day VaR, we combine the total h -day GARCH volatility forecast with the horizon copula to generate scenarios directly. This provides a fast, realistic horizon estimate without requiring thousands of step-by-step daily simulations.

We compare five models for forecasting 99% h -day portfolio risk:

- **Historical simulation** using overlapping h -day returns, and a simple **Gaussian** $\sqrt{h}$ rule (our two baseline benchmarks).
- **Daily copula, $\sqrt{h}$** : takes 1-day VaR from the daily GARCH-EVT- t copula model and scales it up by $\sqrt{h}$. This is the standard shortcut used across the financial industry.
- **Daily copula, h -day vol**: uses the same daily copula, but plugs in the proper full-horizon GARCH volatility instead of scaling by $\sqrt{h}$.
- **Horizon copula (our proposed approach)**: uses the same margins and full-horizon volatility, but fits the Student- t copula directly to the matching horizon view: view 1 (> 4 days) for a 5-day horizon, view 3 (> 16 days) for a 20-day horizon, and view 5 (> 64 days) for a 60-day horizon.

Comparing the second and third models shows the effect of fixing the volatility rule. Comparing the third and fourth shows the effect of fixing the tail dependence. We simulate 20,000 portfolio losses for each model using the exact same random seeds, so any differences reflect true model behavior rather than simulation noise.

2.5. Out-of-sample backtest

At the close of each day t , we forecast portfolio losses over the next h trading days (from $t + 1$ to $t + h$) using only data available up to day t . We re-estimate the models every 21 trading days on a rolling window of the past 1,000 days (roughly four years), updating the daily GARCH volatilities every single day. We run the wavelet filter strictly inside each window so that future data never leaks

into the past. Our backtest begins in March 2011, giving us 3,835 out-of-sample forecast days for each horizon.

We evaluate model performance using standard tests: the Kupiec coverage test [5], the Christofersen independence test [2], and the worst Basel traffic-light zone across any 250-day window (used here as a practical check on breach clustering, following [1]). We also measure overall forecast accuracy using the Fissler-Ziegel (FZ0) joint scoring rule for VaR and Expected Shortfall [3, 7]. The FZ score is strictly consistent: a lower score directly indicates a more accurate forecast that avoids being either recklessly tight or unnecessarily conservative. Because overlapping h -day windows share data, our breach tests evaluate non-overlapping windows (every h -th day), while differences in FZ scores are tested with the Diebold-Mariano test using Newey-West standard errors. Table 2 summarizes our main modeling decisions and the practical reasoning behind each.

Choice	Reason
MODWT, not DWT	Keeps every daily observation at all scales and avoids downsampling; results do not depend on which calendar day we start.
sym8, level 5	Nearly symmetric filter that prevents time shifts; level 5 spans roughly 64 days (one quarter) while preserving enough data for estimation.
Reflection boundary	Mirrors the ends rather than wrapping them around, preventing artificial jumps at the edge of each rolling window.
Cumulative horizon views	A trader holding for h days experiences all slower movements combined, not an isolated frequency band.
Ranks, not GARCH-filtered views	Fitting GARCH on smooth series is unstable; ranks give pure copula margins and are completely reproducible.
$q = 5\%$ co-exceedance	Yields roughly 240 observations in each marginal tail, balancing statistical stability with genuine tail focus.
Block bootstrap, 126 days	Preserves the strong serial correlation of smoothed returns, giving valid confidence intervals.
GJR-GARCH- t + EVT margins	Industry standard: captures asymmetric volatility shocks (leverage effect) and heavy tails at the 99% level.
Summed GARCH variance	Accounts for volatility mean reversion over time, which the rigid $\sqrt{h}$ rule misses.
t copula	Best fit across asset pairs, winning 40 of 42 BIC comparisons.
Rolling 1,000-day window	Four years of daily data: long enough to fit heavy tails reliably, short enough to adapt to changing markets.
FZ0 score and DM test	Simple breach tests lack statistical power at 20 days; the FZ score evaluates every day and tests VaR and ES together.

Table 2: Main modelling choices and why we made them.

3. Results

3.1. Tail dependence across horizons

Table 3 and Figure 2 present our main findings on tail dependence across timescales. Three clear patterns emerge.

Suggestive evidence for equity and credit co-exceedance. Lower-tail co-exceedance between equities (SPY) and high-yield credit (HYG) starts at 0.53 on a daily basis and climbs to 0.77 beyond 64 days, well above the 0.44 level that a normal copula with the same correlation would predict. Out of all 30 pair-horizon comparisons we tested (6 asset pairs $\times$ 5 horizons), this change (+0.24, with a 90% bootstrap confidence interval of +0.10 to +0.37) is the single comparison where the confidence interval stays strictly above zero. This increase holds up whether we use 63-, 126-, or 252-day bootstrap blocks. Still, because this is just one statistically significant result out of 30 tests, we view it as suggestive rather than definitive evidence of widespread horizon-dependent tails. The economic intuition makes good sense: when credit spreads widen, corporate distress takes weeks or months to build. A high-yield bond position that seems only moderately linked to stocks on any single day can become tightly linked during a prolonged downturn.

Suggestive evidence for Treasuries; gold shows no detectable horizon shift. On daily data, equities and long Treasuries rarely crash on the same day (0.05, close to pure independence). Beyond

64 days, joint crash frequency rises to 0.22 (with a confidence interval of 0.01 to 0.34, which crosses zero). Interestingly, all 54 joint crash days at that long horizon come from just two specific market episodes: 23 days in early 2009 and 31 days in 2022, when high inflation caused both stocks and bonds to drop together for months. For gold, joint crashes with equities shift from 0.14 to 0.07, but the confidence interval crosses zero (-0.12 to +0.02), meaning we find no statistically reliable horizon shift.

Portfolio sleeve-level test is not statistically significant. Comparing the risky sleeve (SPY, QQQ, HYG) directly against the hedge sleeve (TLT, GLD) shows joint crash frequency moving from 0.06 at one day to 0.19 beyond 64 days. At every horizon, this stays slightly higher than the Gaussian baseline (0.02 at one day). However, the overall change of +0.13 has a 90% confidence interval of -0.05 to +0.33, which includes zero across all horizons. Because multi-month views contain relatively few separate historical crises, this portfolio-level change is not statistically significant. Upper tails (joint rallies) show no corresponding increase, confirming that shared movements are concentrated in downturns rather than bull markets.

Horizon	Risky vs hedge	Change vs daily	Gaussian	SPY-HYG	SPY-TLT	SPY-GLD	SPY-QQQ	TLT-HYG	TLT-GLD
Daily	0.06 [0.04, 0.09]	–	0.02	0.53	0.05	0.14	0.75	0.12	0.14
> 4d	0.09 [0.05, 0.13]	+0.02 [+0.00, +0.07]	0.03	0.55	0.07	0.17	0.74	0.13	0.14
> 8d	0.11 [0.05, 0.17]	+0.05 [+0.00, +0.11]	0.03	0.50	0.09	0.14	0.76	0.14	0.12
> 16d	0.10 [0.06, 0.16]	+0.04 [+0.00, +0.10]	0.04	0.55	0.05	0.12	0.76	0.08	0.10
> 32d	0.15 [0.01, 0.27]	+0.09 [-0.05, +0.22]	0.04	0.63	0.08	0.06	0.72	0.14	0.11
> 64d	0.19 [0.00, 0.38]	+0.13 [-0.05, +0.33]	0.04	0.77	0.22	0.07	0.62	0.19	0.22

Table 3: Lower-tail co-exceedance at the 5% level by horizon view. Brackets are 90% moving block bootstrap intervals.

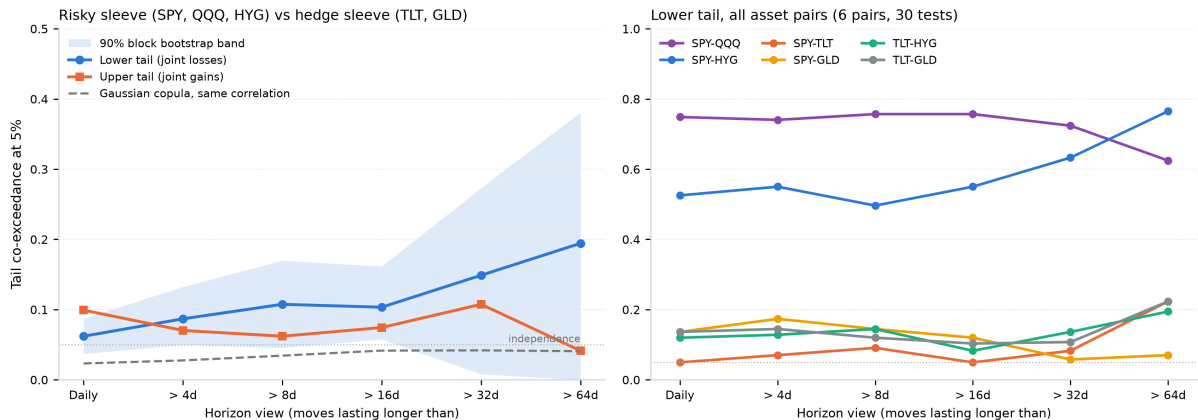


Figure 2: Left: lower and upper tail co-exceedance of the risky and hedge sleeves by horizon view, with the 90% block bootstrap band and the Gaussian copula benchmark. Right: lower-tail co-exceedance for all six asset pairs across the 30 comparisons. The dotted line is independence.

3.2. What ignoring it does to measured risk

Figure 3 compares 99% VaR forecasts between the horizon copula and the daily copula across the entire backtest. Because both models use the exact same margins and full-horizon GARCH volatility, the difference comes entirely from the dependence structure (linear correlation shifts and tail thickness combined). Fitting the copula to the matching horizon increases VaR by 2.1% at 5 days, 5.0% at 20 days, and 4.3% at 60 days on average. This impact varies over time: at 20 days, the spread between the 10th and 90th percentiles ranges from -1.0% to +10.9%, and at 60 days from -10.7% to +15.9%, reflecting that longer horizons have fewer independent market cycles.

In daily practice, the traditional $\sqrt{h}$ rule easily masks this underestimation. Scaling 1-day VaR by $\sqrt{h}$ adds 8.7% more VaR at 20 days than the true multi-day GARCH volatility, and 18.2% more at 60 days. This happens because spikes in market volatility usually subside over time (mean reversion), while the $\sqrt{h}$ formula assumes today's high volatility will persist unabated. Because this volatility

inflation is larger than the copula understatement, the $\sqrt{h}$ rule ends up with enough capital purely by accident. A trading desk that fixes its volatility forecasting while keeping a daily copula would strip away this accidental cushion, leaving the portfolio under-capitalized. (A quick note on how these numbers are averaged: the 8.7% gap is the average daily percentage difference between the two models, while the 4.7% capital savings is calculated by comparing the overall average VaR numbers across the full period, 6.8% vs. 7.1%.)

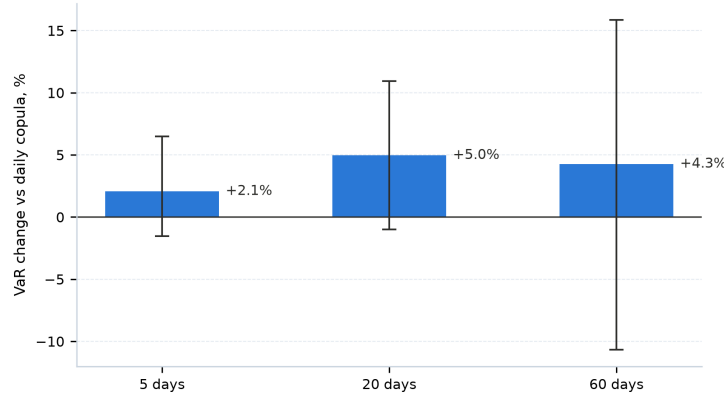


Figure 3: Average change in VaR from using the horizon copula instead of the daily copula, with the same margins and volatility. Bars are means over all forecast days; whiskers are the 10th and 90th percentiles.

3.3. Out-of-sample backtest

Table 4 shows the out-of-sample backtest results. At the 1-day horizon, all three copula models make identical forecasts. They register 37 breaches compared to 38 expected, easily passing both the Kupiec coverage test ($p = 0.83$) and the Christoffersen independence test ($p = 0.40$), reaching the yellow zone at worst during the volatile months after COVID-19. Meanwhile, both naive benchmarks fail: the Gaussian rule suffers 68 breaches, and historical simulation clusters its 41 breaches, pushing both into the red zone.

At the 20-day horizon, the horizon copula achieves the lowest (best) point-estimate FZ score among all five models, recording 2 breaches in 191 non-overlapping windows against 1.9 expected. However, this score improvement is not statistically significant ($p = 0.32$ against the daily copula, and $p = 0.25$ against $\sqrt{h}$). With only 191 non-overlapping windows, simple breach counts do not have enough statistical power to distinguish between good models. At 5 days, the daily $\sqrt{h}$ model actually scores slightly better, with all three copula models scoring within statistical noise of one another ($p = 0.51$).

During the 2022 rate-hiking cycle, every model struggled. Stocks and bonds dropped simultaneously for most of the year. Across 209 forecast days at the 20-day horizon, all copula models experienced severe breach clustering compared to roughly 2 expected breaches: the daily copula suffered 16 breaches, while the horizon copula recorded 13 (saving just 3 bad days). The $\sqrt{h}$ rule recorded 11 breaches, while requiring substantially higher average capital over the full backtest. During the initial COVID-19 crash, all three copula models recorded 5 breaches, as the shock hit too fast for any 20-day model to adjust in advance. Because multi-day forecasts overlap, these crisis counts serve as practical diagnostics rather than formal statistical tests.

h	Model	Breaches / exp.	Kupiec p	Christ. p	FZ	ΔFZ (p)	Avg VaR	Basel
1d	Historical simulation	41 / 38.4	0.67	0.00	-3.612	+0.308 (0.05)	1.75%	Red
1d	Gaussian, $\sqrt{h}$	68 / 38.4	0.00	0.00	-3.381	+0.539 (0.06)	1.47%	Red
1d	All three copula models	37 / 38.4	0.83	0.40	-3.920	–	1.60%	Yellow
5d	Historical simulation	6 / 7.7	0.53	0.76	-2.686	+0.271 (0.23)	3.92%	–
5d	Gaussian, $\sqrt{h}$	10 / 7.7	0.42	0.11	-2.379	+0.578 (0.17)	3.19%	–
5d	Daily copula, $\sqrt{h}$	10 / 7.7	0.42	0.61	-2.973	-0.015 (0.19)	3.58%	–
5d	Daily copula, h-day vol	10 / 7.7	0.42	0.61	-2.957	–	3.45%	–
5d	Horizon copula (proposed)	10 / 7.7	0.42	0.61	-2.966	-0.008 (0.51)	3.52%	–
20d	Historical simulation	3 / 1.9	0.46	0.76	-1.804	+0.432 (0.18)	7.33%	–
20d	Gaussian, $\sqrt{h}$	2 / 1.9	0.95	0.84	-1.861	+0.376 (0.27)	6.03%	–
20d	Daily copula, $\sqrt{h}$	1 / 1.9	0.47	0.92	-2.233	+0.003 (0.95)	7.15%	–
20d	Daily copula, h-day vol	2 / 1.9	0.95	0.84	-2.237	–	6.49%	–
20d	Horizon copula (proposed)	2 / 1.9	0.95	0.84	-2.271	-0.034 (0.32)	6.81%	–

Table 4: Rolling out-of-sample backtest of 99% VaR and ES. Coverage tests use non-overlapping windows. FZ is the Fissler-Ziegel score (lower is better); ΔFZ is the difference to the baseline model (Daily copula, h-day vol) with its Diebold-Mariano p -value.

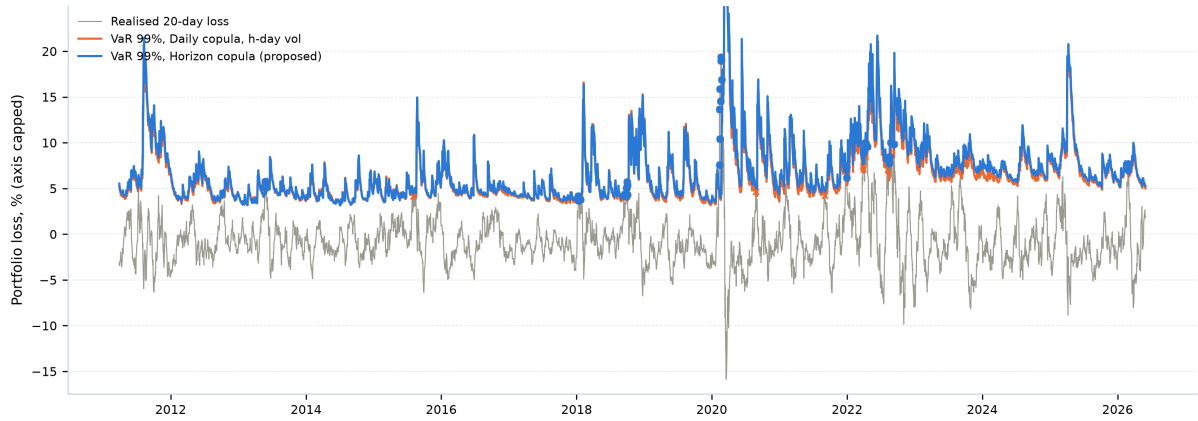


Figure 4: Realised 20-day portfolio loss against the 99% VaR of the daily copula and the horizon copula, March 2011 to June 2026. Markers show breaches. The axis is capped so the March 2020 peak does not flatten the rest.

4. Recommendation

For positions held 20 trading days or longer, prefer a full-horizon GARCH volatility forecast combined with a copula fitted to the matching wavelet horizon view rather than relying solely on the $\sqrt{h}$ rule.

The out-of-sample scoring difference is not statistically significant, so this should be viewed as a model-structure improvement and monitoring framework rather than proof of superior forecasting accuracy.

Why. For our portfolio, making this change frees up 4.7% in regulatory capital on average compared to the $\sqrt{h}$ rule (6.8% versus 7.1% of portfolio value), while achieving the best point-estimate FZ score among all five models and directly aligning model assumptions with the investment horizon. The $\sqrt{h}$ rule overstates 20-day risk on average but understates long-term asset dependence; the first mistake simply cancels out the second most of the time. But this cancellation breaks down when market volatility is unusually low, or during severe structural shifts like 2022 where both asset classes sell off together.

How to put it into practice tomorrow. A risk desk can implement this change directly on top of an existing GARCH-copula system:

1. Keep your standard daily GARCH-EVT margins. For an h -day horizon, sum the daily GARCH variance forecasts for days $1, \dots, h$ instead of multiplying 1-day variance by h .

2. Once a month, run the MODWT on the past four years of returns, extract the horizon view that matches your target holding period (view 3, movements longer than 16 days, for a 20-day horizon), and refit the copula to those ranks.
3. Simulate h -day scenario losses using the new copula and the multi-day margins.

All three steps are implemented in `src/horizon_var.py`. The monthly refit takes just a few seconds of compute time.

When to use it. A trading desk should apply horizon-matched copulas when holding positions for $h \geq 20$ trading days in portfolios containing credit, spread products, or assets whose correlations tighten over weeks. For short holding periods ($h \leq 5$ days) or simple large-cap equity portfolios, stick with standard daily copulas or $\sqrt{h}$ scaling. At short horizons, our backtest shows no statistically meaningful improvement ($p = 0.51$), and standard methods provide solid protection at lower operational complexity.

What to monitor. Track the horizon-matched VaR alongside daily-copula VaR. When the spread between them widens, as it did during 2021 to 2023 (9.7%, 7.9%, and 10.1% by year), it signals that portfolio hedges are losing effectiveness over multi-week horizons. Risk managers should treat this widening gap as an early prompt to review hedge quality rather than simply holding more capital.

5. Limitations

No model is without limitations, and ours has several:

- Our longest horizon views contain relatively few separate crash episodes. This explains why the sleeve-level test and 60-day risk estimates have wide confidence intervals. Longer historical data would help, but high-yield ETF history only begins in 2007.
- Because the backtest requires a 1,000-day rolling window to train, our out-of-sample evaluation starts in March 2011. The 2008 financial crisis is used to train the initial models and is not evaluated out of sample.
- Evaluating 20-day forecasts yields only 191 non-overlapping windows across our test period. Simple breach tests lack the statistical power to reliably tell models apart on such small counts. We rely on the continuous FZ score, which scores every forecast, though score differences between top models remain within statistical noise ($p = 0.32$ at 20 days; at 5 days daily $\sqrt{h}$ scores best).
- We fit the horizon copula directly to the ranks of smoothed returns without filtering out volatility clustering first. Shared volatility spikes may increase measured dependence. The daily copula is fitted the exact same way, so the comparison between them is fair, but readers should keep this in mind.
- Our multi-day VaR scales daily EVT tail shapes by the multi-day volatility forecast. Because adding returns over many weeks tends to smooth extreme tails toward normality (by the central limit theorem), using daily tail parameters may slightly overstate tail heaviness at longer holding periods.
- We approximate multi-day portfolio risk by combining cumulative horizon volatility with a horizon copula, rather than running a full step-by-step path simulation that updates daily GARCH states sequentially.
- The measured difference in VaR between daily and horizon copulas captures the combined effect of horizon dependence (correlation changes and copula tail thickness together), rather than isolating tail dependence by itself.
- Portfolio weights are fixed with no rebalancing costs, bid-ask spreads, or liquidity constraints.

6. Conclusion

Evidence that tail dependence changes with the investment horizon is suggestive for equities and high-yield credit (the only test out of 30 where the confidence interval stays clear of zero), suggestive for equities and Treasuries, and undetectable for gold or the portfolio as a whole. A risk model fitted

to the matching horizon produces roughly 5.0% higher 20-day VaR on average than a daily copula, and over 10.9% higher on one day in ten. However, out-of-sample evidence that this higher estimate performs better is weak ($p = 0.32$). The common $\sqrt{h}$ rule hides this dynamic by overstating 20-day VaR across the board. Replacing $\sqrt{h}$ with full-horizon volatility and horizon copulas addresses both problems, achieves the best point-estimate score at 20 days, and frees up capital compared to the rule it replaces.

Reproducibility and team declaration

Running `python run_all.py` rebuilds every number, table, and figure in this report from local price data in about six minutes, producing identical results across runs. All numbers cited in this text are exported by the script directly to `tables/key_numbers.tex`. Software versions are pinned in `requirements.txt` (Python 3.12), and the test suite passes under `pytest`.

We used AI tools to assist with coding, debugging, and drafting text. Our team chose the research question, data, methods, and interpretations, verified every result against the underlying code, and takes full responsibility for the entire submission. A detailed disclosure is provided in `docs/AI_DISCLOSURE.md`.

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